

Title:

Are you at a risk of getting heart disease?

Basic Idea:

We will take a dataset about heart disease (linked below) and analyze each of the attributes given related to heart disease, including age, gender, cholesterol levels, etc. We will then graph and plot them in different ways (explained below). Link: <https://www.kaggle.com/ronitf/heart-disease-uci>

What will output to the console?

A possible output to the console could be a correlation map showing how closely related each variable is, multiple graphs (likely count plots, line plots, bar plots, and box plots) showing the prevalence of heart attack by the different variables (attributes above), and a final graph showing how effective certain prediction models are at predicting the data we have. From our initial research, some of these prediction models could include a Regression or a TreeClassifier.

What will the user be able to do?

Not much. This is not meant to be really interactive, rather it is meant for the user to look at the different plots and understand what each variable means in the context of the data we are given. We want them to (and we will do this as well) make some statements about heart disease with respect to the dataset. For example, there is a higher chance for a heart attack at an older age.

CV Credit:

For regular credit on our assignment, we will make the correlation map/plot to show the relation between the variables and make the graphs that shows the prevalence of heart attack by the different variables. Making these graphs also means cleaning through the dataset for any bad values and doing some preprocessing (may take some research). Finally, to make the project look more interesting, instead of just having graphs output to the console, we can put together a small written portion on the side that can include our initial hypotheses, graphs to support or refute our hypotheses, and a final statement to link heart disease to the attributes.

For the 1st CV credit, we will research two prediction models used to accurately predict data and incorporate them into our dataset. We will show it's accuracy via another graph to compare which model is better.

For the 2nd CV credit, we will research two more prediction models for data modelling. Similar to the 1st CV credit, we will incorporate these models and graph their accuracy on a plot to compare them.

For the last CV credit, unless we think of another possible addition to our project, we will work on a separate option individually.